

Fifth Semester
Name of Paper: Software Engineering-I
Course No.: BCA/5/CC/25

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **5**
(4- 1- 0)

Unit-1:

(10 L)

Software Engineering: Definition, Software Process, Software characteristics, The changing Nature Of Software.

Software Life Cycle Models: Build and Fix Model, Waterfall Models, Increment Process Models: Iterative enhancement Model, Rapid Application Development (RAD) Model, Evolutionary Process Models: Prototyping Model, Spiral Model.

Unit-2:

(10 L)

Software Requirements analysis & specifications: Requirements engineering, Types, Feasibility Studies. Software Prototyping. Requirement elicitation techniques like Interviews, Brainstorming, FAST, QFD, Use case approach. Requirements analysis using DFD, Data dictionaries & ER Diagrams, Requirements documentation Nature of SRS, Characteristics & organization of SRS. Requirement Validation & Management.

Unit- 3:

(10 L)

Software Design: Definition and objectives. Modularity: Coupling & Cohesion. Strategy of Design: Hybrid, Bottom-Up & Top-Down. Function Oriented and Object Oriented Design.

Unit-4:

(10 L)

Software Metrics: Definition. Token Count, Data Structure Metrics, Information Flow Metrics.

Software Project Planning: Size Estimation: LOC & Function Count, Cost Estimation: COCOMO. Software Risk Management.

Unit-5:

(10 L)

Software Testing: Definition and software Testing Fundamental, Verification and Validation, Alpha, beta and Acceptance Testing, Black box testing, White Box testing, Levels of testing, Debugging.

Software Reliability: Basic Concept, Software and Hardware Reliability, Failure and Faults, Software Quality Attributes.

Software Maintenance: Definition, Categories of maintenance, Maintenance Process. Reverse Engineering, Software Re-engineering, Documentation.

Recommended Books:

1. R. Pressman: *Software Engineering*, McGraw-Hill Science/Engineering/Math; 7th edition (2009)
2. K. K. Agarwal and Y. Singh: *Software Engineering*, New Age International (2005)

Fifth Semester
Name of Paper: Computer Graphics
Course No.: BCA/5/CC/26

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **4**
(3- 1- 0)

Unit-1:

Introduction: History of computer graphic, History of Computer Graphics, What is CG, Types of Computer Graphics, Area of Computer Graphics, **Overview of Graphics Systems:** Video Display Devices, Refresh Cathode Ray Tubes, Raster-Scan and Random-Scan Systems, Input Devices, Hard-Copy Devices and Graphics Software.

Unit 2:

Output Primitives: Points, Line Drawing Algorithms (DDA and Bresenham's Line Drawing Algorithm), Circle- Generating Algorithms (Bresenham's and Midpoint Circle Algorithms), Ellipse-Generating Algorithms (Midpoint Ellipse Algorithm only), Filled- Area Primitives: Scan-Line Polygon Fill Algorithm, Boundary-Fill Algorithm, Flood-Fill Algorithm.

Unit 3:

Two Dimensional Geometric Transformations: Basic Transformations, Matrix Representations and Homogeneous Coordinates, Composite Transformations, Reflection and Shear, Transformations between Coordinates Systems, Raster Methods for Transformations.

Two-Dimensional Viewing: The Viewing Pipeline, Viewing Coordinate Reference Frame, Window-to-View Port Coordinate Transformation, Clipping- Point, Line (Cohen-Sutherland Line Clipping and Liang –Barsky Line Clipping and Nicholl-Lee-Nicholl Line Clipping) and Polygon Clipping (Sutherland-Hodgeman Polygon Clipping, Weiler-Atherton Polygon Clipping).

Unit 4:

Three Dimensional Geometric Transformations: Translation, Rotation, Scaling, Reflection and Shears, Composite Transformations, Modeling and Coordinate Transformations.

Three Dimensional Viewing: Viewing Pipeline, Viewing Coordinates, Projections and Clipping.

Unit 5:

Multimedia Systems Design: Multimedia Elements, Multimedia Applications, Multimedia System Architecture, Evolving Technologies for Multimedia Systems, Multimedia Data Interface Standards, the Need for Data Compressions, Multimedia Database.

Media and Data Streams: Medium, Main Properties of a Multimedia Stream, Multimedia System Definition, Combination of Media.

Data & File Format Standards: Rich –Text Format, TIFF File Format, RIFF, MIDI File Format, JPEG DIB File Format, MPEG Standards.

Reco0mmended Books:

1. D. Hearn & M. P. Baker - *Computer Graphics C Version*, 2nd Edn, Pearson Education, New Delhi, 2006
2. J. F. Koegel Buferd -*Multimedia Systems*, Pearson Education, New Delhi, 2006

Fifth Semester
Name of Paper: GUI Programming
Course No.: BCA/5/CC/27

Marks Scale: **100** marks (End Sem. Exam: 75 + Internal: 25)

Credit: **4**
(3 - 1 - 0)

Unit-1:

(10L)

An overview of the .NET framework, Common Language Runtime (CLR), The .Net Framework Class Library, Code Loading and Execution, Common Type System (CTS), Common Language Specification (CLS), MSIL. Introduction to .NET Architecture, Event-Driven Programming, Integrated Development Environment (IDE) for VB.Net, Introduction to VB.NET – basic syntax, Identifiers, Keywords, Data types, Type conversion, Variables, Constants and Enumerations, Statements, Operators.

Unit-2:

(10 L)

Working with simple applications and complex applications, **Working with Forms:** Textbox, Label, Button, ListBox, ComboBox, CheckBox, PictureBox, RadioButton, Panel, Timer, ListView, TreeView, DateTimePicker, LinkLabel, Methods and Events, Event Handling.

Dialog Boxes: OpenFileDialog, SaveFileDialog, FontDialog, ColorDialog, PrintDialog.

Designing menus: MenuStrip, ContextMenuStrip, Access & Shortcut keys.

Unit-3:

(10 L)

Decision Making: Looping statements, Strings, Arrays, VB.NET – Collections.

Procedures and Functions: Defining a function, Function returning a value, Recursive function, Param Arrays, Passing Arrays as functions arguments, Defining Sub Procedures, Passing parameters by Value, Passing parameters by Reference.

Major Error Types: Syntax, Execution and Logic errors. Exception, Exception handling, ureating user-defined Exceptions.

Unit-4:

(10 L)

Introduction to object oriented programming, Classes and Objects, Defining a Class, Inheritance, Methods and Properties, Method Overloading and Method Overriding, Constructors and Destructor, Interface.

Access modifiers: Public, Private, Protected, Friend. Using namespace, using imports statement, creating class library.

Unit-5:

(10 L)

Introduction to database access, overview of ado.net, ado.net architectures and its components, Data Provider: Command, Connection, Data Reader, Data Adapters, Data Set: Data Table Collection, Data Relation Collection, Extended Properties, Data Tables, Data Relation, Data Row Collection, Data View, Primary Key, Data Columns, Data Rows, Data Column Collection, Connecting to a Database, Working with Data Grid View, Reporting using Report Viewer, Data binding with textbox, data grid, Binding Source Navigator.

Recommended Books:

1. Thearon Willis, Bryan Newsome: *Beginning Microsoft Visual Basic 2010*, Wiley India Pvt. Ltd. (2012).
2. Evangelos Petroustos: *Mastering Microsoft Visual Basic 2010*, Wiley India Pvt. Ltd. (2010).
3. http://www.tutorialspoint.com/vb.net/vb.net_quick_guide.htm

Fifth Semester
Name of Paper: Elective-I (any one)
Course No.: BCA/5/EC/28
a) Cloud Computing

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **5**

(4- 1- 0)

Unit – 1:

Understanding Cloud Computing: Origin and Influences, Basic Concepts and Terminology, Goals and Benefits, Risks and Challenges.

Unit – 2:

Fundamental Concepts and Models: Roles and Boundaries, Cloud characteristics, Cloud Delivery Methods, Cloud Deployment Models.

Unit – 3:

Cloud-Enabling Technology: Broadband Networks and Internet Architecture, Data Center Technology, Virtualization Technology, Web Technology, Multitenant Technology, Service Technology.

Unit – 4:

Fundamental Cloud Security: Basic Terms and Concepts, Threat Agents, Cloud Security Threats.

Unit – 5:

Cloud Infrastructure Mechanisms: Logical Network Perimeter, Virtual Server, Cloud Storage Device, Cloud Usage Monitor, Resource Replication.

Recommended Books:

1. Thomas Erl: *Cloud Computing: Concepts, Technology & Architecture, 1st Edition*, Pearson (2013).
2. Raj Kumar Buyya, James Broberg, Andrzej M. Goscinski: *Cloud Computing: Principles and Paradigm*, Wiley(2013).
3. Dinkar Sitaram, Geetha Manjunath: *Moving to The Cloud*, Elsevier (2012)

Fifth Semester
Name of Paper: Elective-I (any one)
Course No.: BCA/5/EC/28
b)E-commerce and E-governance

Marks Scale: 100 marks (End Sem. Exam: 75 + Int.: 25)

Credit: **5**

(4- 1- 0)

Unit-1:

(10 L)

Internet and Commerce: Business operations; E-commerce practice vs. traditional business practice; Concepts b2b, b2c, c2c, b2g, g2h, g2c; Benefits of e-commerce; Limitation of E-commerce Management issues relating to e-commerce.

Operations of E-commerce: Credit card transaction; Secure Hypertext Transfer Protocol (SHITP); Electronic Payment System; Secure Electronic Transaction (SET); SET's encryption; Process; Cybercash; Smart Cards; Indian Payment models.

Unit-2: Applications in B2C

(10L)

Consumers' shopping procedure on the internet; Impact on dis-intermediation and re-intermediation; Global market; Strategy of traditional department stores; Products in b2c model; Success factors of e-brokers; Broker-based services online; Online travel tourism services; Benefits and Impact of e-commerce on travel industry; real estate market; Online stock trading and its benefits; Online banking and its benefits; Online financial services and their future; E-auctions-benefits, implementation, and impact.

Unit-3: Applications in B2B

(10L)

Applications in B2B: Application of b2b; Key technologies for b2b; Architectural models of b2b; Characteristics of the supplier-oriented marketplace, buyer-oriented marketplace, and intermediary-oriented marketplace; Benefits of b2b on procurement reengineering; Just in Time delivery in b2b; Internet-based EDI from traditional EDI; Integrating EC with back-end information systems; marketing issues in b2b.

Unit 4: Introduction to e-Governance

(10 L)

E-Governance: Needs of E-Governance, Issues in E-Governance applications and the Digital Divide; Evolution of E-Governance, Its scope and content, components of e-Governance, Present global trends of growth in E-Governance, Areas of e-Government, Benefits of E-Governance, E-Governance Vrs E-Government, Critical success and failure factors for e-Governance. Role of social Media in e-Governance.

Unit 5: E-Governance Models

(10L)

Introduction, **Models of Digital Governance:** Broadcasting Dissemination Model, Critical Flow Model, Comparative Analysis Model, Mobilization and Lobby Model, Interactive-Service Model, E-governance Maturity Model, principle, applications and evaluation of each model. Characteristics of Maturity levels, Key Focus areas, towards good Governance through E-Governance models

Recommended Books:

1. Efraim Turban, Jae Lee, David King, H.Michael Chung, *Electronic Commerce–A Managerial Perspective*, Addison-Wesley.
2. C.S.R. Prabhu: *E-Governance: Concepts and Case Studies*, Prentice-Hall of India Pvt. Ltd. (2004).

Fifth Semester
Name of Paper: Elective-I (any one)
Course No.: BCA/5/EC/28
c) IT Act and Cyber Laws

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **5**

(4- 1- 0)

Unit-1:

(10 L)

Basic Concepts of Technology and Law : Definition Cyber Law, cyber law: Cyber-crimes, electronic & Digital Signature, Intellectual property, Data protection and privacy, Scope and needs of Cyber Laws, The jurisprudence of Indian Cyber Law.

Unit-2:

(10 L)

Evolution of cyber-crime, Cyber Fraud and Cyber Cheating, Virus on the Internet, Email spoofing, E-mail bombing, cyber stalking, Denial of service attacks, cyber Terrorism, Salami attack, Online gambling, Sale of illegal articles, Internet time theft, Web jacking, Data diddling, Intellectual Property crimes, Web defamation, Cyber Pornography.

Unit-3:

(10 L)

Law of Digital Contracts: The essence of Digital Contracts, The system of Digital signatures, Digital Signature Certificates, Certifying Authorities and Liabilities, The role and function of certifying authority.

Unit-4:

(10 L)

E-Governance and IT Act 2000 & Amendments: Legal recognition of electronic records, Legal recognition of digital signature, Use of electronic records and digital signatures in Government and its agencies.

Information technology Act 2000: Object and Scope of the IT Act: Genesis, Object, and Scope of the Act. Major issues addressed by the IT Act, Extent and jurisdiction of IT Act, Applicability of IT Act, and Relevant Authorities in India.

Unit 5:

(10 L)

Copyright: Meaning, Ownership and Assignment, Licence of Copyright, Copyright Protection of Content on the Internet.

Management Issues: Organizational Issues

Introduction, Cyber law: Management issues, Cyber law: Organizational issues, Jurisdictional issues, Online Dispute Resolution (ODR)

Recommended Books:

1. Farooq Ahmad: *Cyber Law in India*- Pioneer Books.
2. Vivek Sood: *Cyber Law Simplified*, Tata McGraw Hill.
3. Vakul Sharma: *Information Technology Law and Practice*- Universal Law Pub. Co. P. Ltd.
4. Suresh T Vishwanathan: *The Indian Cyber Law* –Bharat Law House New Delhi.
5. The Information Technology Act, 2000 – Bare Act –Professional Book Publishers – New Delhi.

Fifth Semester
Course No.: BCA/5/CC/29
Name of Paper: Minor Project

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **3**
(0-0-3)

Introduction

The BCA students are encouraged to involve themselves on this minor project. It is advised to students to develop their own project for solving different types application developments. Doing this will give more exposure to handle real life problems of project development. The hands on experience gained from the practical classes provide students the knowledge to work with software developments in variety of application domains.

From the beginning of selecting a topic to completing this minor project, the teacher in-charge is expected to give guidance to student wherever necessary. The teacher is expected to explain and refresh the minds of students regarding different topics relation to project development such as Software Requirements Specifications, Entity Relationships diagrams, Data Flow Diagrams, components and procedure for writing project report, etc. But majority of the minor project classes will be devoted for the students to do their practical project works.

The primary emphasis of the minor project work is to understand and gain the knowledge of the principles of software engineering practices, and develops good understanding of Software Development Life Cycle. Students should take this project work very seriously. Topics selected, should be not too simple and but easy enough for the standard of BCA student. The project should be genuine and original in nature and should not be copied from anywhere else. Student must follow the different guidelines given for this minor project.

Marks Distribution

The total mark for this minor project is 100 which is divided as 75 external marks and 25 internal marks. Marks distribution for internal as well as external for this minor project is given below:

SINo.	Topics	Marks
1	Project Development and Execution	30
2	Project Report	15
3	Presentation	20
4	Viva Voce	10
5	Internal mark (including attendance)	25
TOTAL		100

Note: The internal marks 25 includes marks to be given based on attendance or regularity of the class and any other methods of assessments such as test, presentations, etc. can be adopted by the teacher for rewarding the internal mark.

Grouping

This is the first project work for BCA students and therefore each individual student is expected to carry out their own Minor project work which is not too complex but easy enough to be able complete a student. But the minor project should be not too simple and must be of at the level of BCA standard. Under certain cases, if students want to work in

group, *maximum of 3(three) students will be allowed to work in a single project*. But each student must take part in all the different phases this minor project development. Any member of the group, not taking part in the project development may be rewarded negatively.

Schedule

The teachers must give time frame or calendar for the minor project. This should be done after completing theoretical explanation of the minor project development process including SRS, SDLC, etc. The time frame or calendar for minor project should be display on notice and websites by the teacher concern. The format of the calendar for minor project is given below:

SINo	Topics	Date / Last Date
1	Submission of Project Synopsis	
2	Approval of Project Proposal	
3	Submission of Project Report	
4	Project Presentation and Viva-voce	

Project Synopsis

Students must prepare project synopsis before carrying out the actual project work. The project synopsis should be submitted to the teacher and only obtaining approval from the teacher, actual work on the project can be carry on. This will avoid any redundancy in selecting the project work. Changing a project after approval can be done only in consultation with the project guide by preparing a fresh project synopsis.

The format for Project Synopsis must include the following:

- (1) Title of the project
- (2) Introduction and Objective of the project
- (3) Software Requirements for the development of the project
- (4) Students' details with signature

Project Report Format

Project report should be printed on a single sided A4 page with proper line spacing such as 1.5pt and the report must include the followings:

- 1) Cover page
- 2) Certificate
- 3) Acknowledgement
- 4) Contents
- 5) Introduction
- 6) Objectives
- 7) Need for system
- 8) Feasibility study
- 9) Hardware requirement for running the software
- 10) Software requirement for running the software
- 11) Entity Relationship Diagrams
- 12) Data Flow Diagrams
- 13) Input and Output Screens along with codes
- 14) Limitations of the Project
- 15) Future Scope of the Project
- 16) Bibliography

All the students must submit project report in his/her name only to the teacher and sharing of project report is not allowed. Minor project report may be spiral bind or hard bind. Besides submitting to the department, each student is expected to retain another copy of the project report for their future references.

Format of Certificate

The format of Certificate for the project is given below.

Name of the College or Institute

Institute
Logo

CERTIFICATE

This is to certify that these practical records of the course “**Minor Project**” with course code **BCA/5/CC/29** submitted for the partial fulfillment of the requirement **BCA _____ Semester Examination** is an authentic and original work carried out under my supervision. Detail of student is given below:

University RollNo. _____

University RegNo. _____

(Name of Student)

BCA Semester : _____

Date : _____

Round
Seal

(Name of Teacher)

Designation

Date : _____

Fifth Semester
Course No.: BCA/5/CC/30
Name of Paper: Programming with VB.NET Lab

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **3**
(0- 0- 3)

A. Record Book
B. Viva Voce
C. Practical

(10 marks)
(15 marks)
(50 marks)

1. Develop a simple windows application that can say "Hello Mr/Ms<yourname>, Welcome to Visual Basic 2010 Programming!" when the OK button is click.
2. Write windows application with three input boxes - two for inputting numbers and one for displaying result. Perform four basic mathematical operating such as plus, minus, multiply and divide using their respective buttons.
3. Develop console application for calculating simple interest where Simple Interest = (Principal x Time x Rate) / 100. Calculate the simple interest at least up to three decimal accuracy.
4. Develop a console application to calculate largest and smallest number in a given array.
5. Write a window application for login purpose with two input boxes for username and password and appropriate buttons. When the user click login button, the software will check the password with the password set by you and then display message box accordingly.
6. Develop windows application having one combo box at the middle with at least 5 color names in it. When user selects any color from the list, the background color of the application has to change as per the selection using the select...case statement.
7. Design a form having 1 Text box and buttons like Bold, Italic, Red & Blue, show and hide When user presses any of this command buttons then the content of text box will be changed to red or blue or bold or italic and Clicking on the "Show" button it will display a Textbox & when user clicks on "Hide" button then it will remove the Textbox.
8. Design a digital watch using Timer Control.
9. Write a birthday program that allow the user to enter their name, select their birthday from the calendar. Display the users name, age and the day when his/her birthday will come this year when the control is clicked.
10. Design a form that demonstrate check box and radio button.
11. Write a VB.Net program to design the following form, select the question number from combo box that question will be displayed into textbox and the options for that question will be displayed on four radio buttons, select option and click on submit button result should be displayed in message box.
12. Write a VB.Net program having two textboxes, accept a character from keyboard in the textbox and check whether it is vowel or not and display the case of that character in message box. And in the next textbox, check if a text entered is an email address or not.
13. Design a calculator program using control array.

14. Implement a program for the demonstration of Array in Visual Basic 2010 where sorting array can be performed by the user.
15. Create an application for the demonstration of different types of dialog controls such as Save Dialog, Open Dialog, Font Dialog, Color Dialog, Folder Browser Dialog and Print Dialog.
16. Add status bar control to any of your existing application and display date and time on the status bar.
17. Write a VB program for error handling.
18. Write VB.Net application for students database with the fields (id, sname, rollno, year, department, father's name, phone no, address, passport size). Create add, delete, edit, previous and next button to navigate the record.
19. Print all the students detail, students by department, students by year using reportviewer.
20. Display the students detail using datagrid view.
22. Use the above table to demonstrate databinding navigator.
23. Write vb.net application for lecturer database with the fields (id, lect name, dob, basic salary, grade pay, da, hra, sca, total pay). Calculate da, hra, sca and total pay from basic salary and grade pay. Save the record in database and display. Use report viewer to view individual record.
24. Add Visual Studio 2010 Setup project to your application and generate single installer and test your installer into your machine and confirm its working.

Note: The above lists of experiments are suggestive of the standard. Department may modify, add or delete any experiment whenever it considered necessary